

Silica Server — Admin Chat Manual

Covers the four DrMuck mods loaded on this server:

Mod	Chat prefix
Si_CrabCannon	/cc
Si_HeavyTeleporter	/st
Si_UnitBalance + UI	!rebalance , !b , /stats
Si_KingOfTheHill	/koh , /buy

All "admin" commands gate on **SilicaAdminMod's** `CanAdminExecute(Power.Generic)` — you need at least the `Generic` power in `admins.json`. Player commands work for everyone.

Most commands save back to their JSON config in `UserData/<ModName>_cfg/` automatically.

Player commands quick reference

Everything in this section works for **any connected player** — no admin powers required. Type these into normal chat; commands with a `/` prefix are hidden from other players (handled as player commands by SilicaAdminMod).

From Si_CrabCannon — set your own super-weapon aim

Available to **commanders** and **regular players** only when admins enable it (via `/cc commanderaim` and `/cc playeraim`). Admins can always use it.

Command	What it does
/cca im	Show your current personal aim (or the cannon + super defaults if you haven't set one)
/cca im <angle> <speed>	Set aim manually. Angle 1–89°, speed 1–800. Range capped at server's <code>MAX_RANGE</code> . Commander = sets super-weapon defaults team-wide and broadcasts to team chat. Player = personal override (only affects your own launches)
/cca im range <meters>	Compute the speed needed to land at <code><meters></code> using your current angle. Convenience for "I want a 1800 m shot" — speed is auto-computed and rejected if it would exceed limits

From Si_HeavyTeleporter — battlefield teleport (Sol & Centauri only)

Command	What it does
<code>/st</code>	Request a heavy vehicle teleport to you. Opens a menu of nearby teleportable vehicles
<code>/st <N></code>	Pick vehicle N from the <code>/st</code> menu (e.g. <code>/st 1</code>)
<code>/st status</code>	Read-only summary: enabled/disabled, your team's tech tier requirement, charges left, cooldown

Requirements: your team's tech tier must be at least the configured threshold, and your team must have at least one charge available. Recharges automatically.

From Si_UnitBalance — inspect any unit's stats

Command	What it does
<code>/stats</code>	Open a read-only inspector for the unit you are currently controlling. Browse parameter groups with <code>/1 – /N</code> . <code>/back</code> goes up, <code>/0</code> closes

You must be controlling a unit. Useful to see actual HP, damage, range, speed numbers (which can differ from vanilla because of live balance overrides).

From Si_KingOfTheHill — capture status + unit purchase

Command	What it does
<code>/koh</code>	Live status — current king team, capture % progress, last tick gain, per-team handicap multipliers
<code>/koh status</code>	Same as above
<code>/buy</code>	Open the unit-purchase menu (modes 1 & 2; falls back to "disabled" in mode 3 or if the server enabled <code>BuyAdminOnly</code>). Categories listed; pick with <code>/1 – /N</code>
<code>/buy <N></code>	Pick category or unit N directly
<code>/1 – /N</code>	Navigate the current <code>/buy</code> menu level
<code>/back</code>	One level up in <code>/buy</code>
<code>/0</code>	Close the <code>/buy</code> menu

Purchased units spawn ~15 m in front of you at ground level. Spawn is refused if it would land inside the configured no-build distance of an enemy HQ / Nest / Queen (configured via `BuyMinDistanceFromEnemyCritical`, typically ~400–800 m).

Credits come from three sources while you play: - **Starter credits** — flat grant once per round (server-configured; often a few thousand) - **Kill rewards** — a percentage of the victim unit / structure's cost goes to the killer - **Zone presence** — if KGT is in full mode (Mode 1) you accrue credits per second while one of your units is inside the capture zone

Si_CrabCannon

Long-range super-weapon for the **Alien faction**. Once the Crab Cannon is unlocked (team reaches the configured tech tier), an alien player walks a Crab (or any enabled creature) into the Nest area and triggers a ballistic launch — the Crab becomes a projectile that arcs across the map, lands at the configured range, and unleashes the creature on the target zone. Combined with the optional **Super Weapon** mode it becomes a per-team strategic strike with charges, recharge time, and team-wide aiming.

Acts as the alien equivalent of an artillery / nuke. Default cannon parameters (speed, angle, payload) are server-wide; the Super Weapon adds per-team charges + a separate, team-aimable trajectory.

All `/cc` subcommands are **admin only**. Players who are allowed to participate use `/ccaim` instead.

Quick reference

Command	What it does
<code>/cc</code>	Toggle ON / OFF
<code>/cc on</code> / <code>/cc off</code>	Explicit on/off
<code>/cc status</code>	One-line summary: mode, speed, angle, peak/range, cooldown, unit toggles, super state
<code>/cc speed <m/s></code>	Launch speed (0–800). Adjusts → previews new peak + range. Rejected if range exceeds map limit
<code>/cc angle <0-90></code>	Launch angle in degrees. Same range guard
<code>/cc range <m></code>	Sets speed so range hits this value at current angle. Convenience for "I want a 1500 m shot"
<code>/cc cooldown <s></code>	Reload between shots
<code>/cc radius <m></code>	Trigger radius around aim point
<code>/cc tier <N></code>	Minimum team tech tier before cannon is available
<code>/cc crab \</code> <code>goliath \</code> <code>behemoth \</code> <code>scorpion \</code> <code>hunter</code>	Toggle that creature type on/off as a payload
<code>/cc playeraim</code>	Toggle: can regular players aim the cannon?
<code>/cc commanderaim</code>	Toggle: can the commander aim the cannon?
<code>/cc countdown <s></code>	Pre-fire warning countdown broadcast in chat

Super weapon (separate sub-system)

Command	What it does
<code>/cc super</code>	Toggle super weapon ON / OFF
<code>/cc superrange <m></code>	Sets the super-weapon's speed for a target range (uses super's angle)
<code>/cc supercharges <N></code>	Max simultaneous super-weapon charges per team
<code>/cc supercd <s></code>	Recharge time between super shots

`/cc status` includes the super line: `SUPER: <used>/<max> charges, cd=<s>, aim=<angle>/<speed>`.

`/ccaim` — companion player command (commanders / players, gated by admin)

Command	What it does
<code>/ccaim</code>	Show personal aim or fall back to cannon + super defaults
<code>/ccaim <angle> <speed></code>	Commander → updates super-weapon defaults team-wide. Regular player → personal aim override. Range checks honor <code>MAX_RANGE</code>
<code>/ccaim range <meters></code>	Computes required speed for target range at current angle

Admins can always use `/ccaim`. Commanders need `CommanderAimAllowed` ON (toggle with `/cc commanderaim`). Regular players need `PlayerAimAllowed` ON (toggle with `/cc playeraim`). Both default to OFF in stock config.

Si_HeavyTeleporter

A **vehicle-teleport tool for human factions** (Sol & Centauri). Once a team reaches the configured tech tier, any soldier on that team can `/st` to call a team vehicle to their current location — useful for getting a Hover Tank or Siege Tank across the map without driving the whole way. Commanders and admins gain extra forms (`/st <playername>`, `/st <x> <z>`) for tactical positioning of teammates or themselves.

Pickup constraint: only vehicles **standing in front of the team's Ultra Heavy Factory** are eligible for teleport (i.e. lined up on the UHF's spawn pad / staging area). The `/st` menu only lists those vehicles.

Each team has a shared charge pool that recharges over time, so it can't be spammed. Min-distance and radius rules prevent teleport-stomping enemy bases.

All player / commander / admin variants of `/st` share the same prefix; access scales with role.

Player commands (anyone, Sol or Centauri only)

Command	What it does
<code>/st</code>	Open a menu listing nearby teleportable vehicles. Next call <code>/st <N></code> picks one
<code>/st <N></code>	Confirm vehicle selection after <code>/st</code>

Requires team tech tier ≥ `TechTier` config and at least one charge.

Commander / admin commands

Command	What it does
<code>/st <playername></code>	Teleport that player's controlled unit
<code>/st <x> <z></code>	Teleport caller to map coordinates (admin or commander)

Admin-only configuration

Command	What it does
<code>/st on / /st off</code>	Master switch
<code>/st status</code>	Live summary + per-team charge/cooldown
<code>/st charges <N></code>	Max charges per team
<code>/st cd <s></code>	Recharge time
<code>/st tier <N></code>	Minimum tech tier required to use teleporter
<code>/st countdown <s></code>	Pre-teleport delay (broadcast warning)
<code>/st radius <m></code>	Teleport target radius (target unit area)
<code>/st mindist <m></code>	Minimum distance from enemy HQ/Nest to allow teleport

Si_UnitBalance

The server's **live unit-balance system** — overrides vanilla numbers (HP, cost, build time, weapon damage, ranges, projectile speeds, movement, FOW radii, etc.) on every Silica unit and structure per faction. The values live in `Si_UnitBalance_Config.json`; the mod applies them as a runtime overlay via the game's OverrideManager without touching prefab assets (so they're cleanly reverted at game end / map change).

The accompanying **UnitBalance UI** layer exposes three chat surfaces: - `!rebalance` — hot-reload the JSON after hand-editing it - `!b` — interactive admin editor: drill into Faction → Category → Unit → Parameter and set a new value live, mid-round - `/stats` — read-only inspector available to any player (lets soldiers see the actual modded numbers for whatever they're piloting)

There's also a built-in **round-transition watchdog** that auto-recovers stalled end-of-round states. Recommended kept ON.

`!rebalance` (admin)

Command	What it does
<code>!rebalance</code>	Re-read <code>Si_UnitBalance_Config.json</code> from disk and re-apply every override (<code>cost_mult</code> , <code>health_mult</code> , <code>range_mult</code> , etc.) to the live game. Takes effect immediately. Use after editing JSON by hand

`!b` — Balance Editor (admin, interactive menu)

Command	What it does
<code>!b</code>	Open the editor. Top-level menu: 1. Sol 2. Centauri 3. Alien 4. JSON 5. HTP
<code>!b <N></code>	Pick item N from current menu without typing <code>!b</code> first then <code>/N</code>
<code>!b exit</code>	Close the editor (also closes on second <code>!b</code> with no args)

Navigate using `/1` – `/20`, `/0` (close), `/back` (one level up). Dotted form also works: `.1`, `.2`, ... and `!b 1 1.5` sets the visible parameter directly.

Menu structure: 1. **Sol / Centauri / Alien** → faction → category (Barracks/Light/Heavy/UHF/Air, or Cysts/Nest) → unit → parameter group (base/primary weapon/secondary weapon/etc) → parameter (set numeric value or 0/1 for bools) 2. **JSON** → 1. Reset to blank (vanilla) · 2. Save current config · 3. Load saved config 3. **HTP** (HitPoint Tier) → Time / Cost split → tech-tier values

 **Conflict with `/buy`** : keep only one of `!b` or `/buy` open at a time. If you have `!b` open and try `/buy`, KGT will refuse: *"You have a balance editor (/b) session open. Close it first (type /b again), then run /buy."*

`/stats` — Read-only inspector (any player)

Command	What it does
<code>/stats</code>	Open the stats view for whatever unit you are currently controlling. Browse parameter groups with <code>/1</code> – <code>/N</code> , <code>/back</code> , <code>/0</code> to close

`/stats` and `!b` are mutually exclusive per player — opening `!b` drops any active `/stats` session automatically.

Watchdog (server safety)

Command	What it does
<code>!bal watchdog_enabled true \ false</code>	Per-server flag. When ON, monitors round-end transitions: if no new round starts within 60 s, forces <code>GameMode.EndRound()</code> so the server self-heals from stalls (e.g. the <code>TrueShadow</code> shader exception bug)

Recommended setting: `true`.

Si_KingOfTheHill (KGT)

The headline **"King of the Galactic Teleport"** game-mode layer. Drops a neutral capture point (an UltraHeavyFactory built by the Wildlife team, configured per map via Si_MapBalance) somewhere on the map. The first team to push enough weighted unit-cap into the capture cylinder and hold dominance for long enough wins the round — even without destroying the enemy HQ. Capture progress is visualized live by a ring of 12 outposts that fill in the leading team's faction colour.

KGT also bundles a **per-player credit economy** (`/buy` menu, kill rewards, zone-presence ticks, faction-treasury commander rewards) and an anti-snowball **handicap multiplier** so a dominant team's commander reward shrinks while a trailing team's grows.

Has two chat surfaces: - `/koh` — admin configuration + live status (status itself is player-visible) - `/buy` — in-round unit purchase menu, available to any player (configurable to admin-only)

A three-tier **Mode system** lets you run the full KGT experience, only the `/buy` economy, or disable everything per round.

Per-map radii: `CaptureRadius` and `ExclusionRadius` (the no-build zone) are normally **defined in the MapBalance spawn config** — the `koh` block of `UserData/Spawns/<MapName>/koh_*.json`. KGT reads them from MapBalance at round start; the matching `Cfg.CaptureRadius` / `Cfg.BuildExclusionRadius` fields in `Si_KingOfTheHill_Config.json` act only as fallbacks when the spawn config doesn't provide a value. **To change a map's capture/no-build radius, edit that map's MapBalance JSON, not the KGT config.**

`/koh` — config + status

`/koh` alone prints the status line. Sub-commands are admin-only.

Command	What it does
<code>/koh</code> or <code>/koh status</code>	One-line summary: ON/OFF, mode, capture radius, exclusion radius, king + progress, last tick gain/cap, per-team handicap multipliers
<code>/koh on</code> / <code>/koh off</code>	Master toggle (<code>Cfg.Enabled</code>)
<code>/koh mode <1\ 2\ 3></code>	Operating mode (see § Mode system)
<code>/koh starter <N></code>	Set <code>StarterCreditsPerPlayer</code> — credits granted to each player once per round (modes 1 & 2 only). <code>0</code> disables
<code>/koh radius <m></code>	Capture-zone radius
<code>/koh threshold <pts></code>	Win threshold (points to win)
<code>/koh rate <pts/s></code>	<code>MaxCapturePointsPerSecond</code> cap
<code>/koh reset</code>	Clear king + accumulation + milestones for the current round (doesn't end the round)

`/buy` — in-round unit purchase

Available in modes 1 & 2. Anyone can use it unless `Cfg.BuyAdminOnly: true` is set.

Command	What it does
<code>/buy</code>	Open buy menu. Shows categories: Barracks, Light Factory, Heavy Factory, Ultra Heavy Factory, Air Factory (humans) — or Lesser/Greater/Grand/Colossal Spawning Cyst (alien)
<code>/buy <N></code>	Pick category N (or unit N once in a sub-list)
<code>/1 – /N</code>	Same as <code>/buy N</code> while the menu is open
<code>/back</code>	Back to category list
<code>/0</code>	Close menu

Units spawn ≈ 15 m in front of your controlled unit at ground level. Refused if too close to any enemy critical (HQ / Nest / Queen) — distance gate is `Cfg.BuyMinDistanceFromEnemyCritical`.

Mode system

`Cfg.Mode` (1 / 2 / 3) — change via `/koh mode N`:

Mode	Intro chat + voice	KoH spawn / capture / immunity	AntiBuild zone	<code>/buy</code>	Kill rewards	Zone-presence reward	Commander reward	Starter credits
1 Full (default)	✓	✓	✓	✓	✓	✓	✓	✓
2 BuyOnly	—	—	—	✓	✓	—	—	✓
3 Disabled	—	—	—	—	—	—	—	—

In **Mode 2**, `MapBalance` is told KGT doesn't want KoH layouts (`IsKohModLoaded()` reflects `ModeKohActive` → false), so the map picker skips `koh_*` JSONs even though KGT is loaded.

Mode changes take effect at the next round start.

How KGT works — mechanics & MapBalance integration

KGT itself does **not** spawn the capture point. `Si_MapBalance` **owns the spawn**; KGT just consumes it.

Spawn pipeline

```
new round starts
↓
Si_MapBalance picks a layout JSON for the map
↓
IF (KGT loaded AND KGT.ModeKohActive)
  → prefers a layout that has a "koh" block (e.g.
  UserData/Spawns/Citadel/koh_test_citadel.json)
  → spawns the koh.PrefabName (e.g. Sol_UltraHeavyFactory) at koh.Position on team koh.Team
  (default Wildlife)
↓
Si_MapBalance fires its public OnSpecialsReady event
↓
KGT.OnMapBalanceSpecialsReady reads MapBalanceSpecials.Koh.{Obj, CaptureRadius,
ExclusionRadius}
↓
KGT caches: _kohTower (transform), _kohCenter (Vector3), _captureRadius, _exclusionRadius,
_kohTeam (Wildlife)
↓
KGT spawns 12 outpost ring markers around _kohCenter at _captureRadius
KGT arms damage immunity (ApplyDamage / SetHealth / SetHealth01 /
OnReceiveClientDamageHitPacket transpiler)
KGT arms AntiBuild via SilicaAdminMod.Event_Construction.OnRequestBuildStructure
KGT arms auto-respawn watchdog (3 s interval – rebuilds the KoH if a player slips damage past
immunity)
```

If the active spawn-config JSON for the current map has **no koh block**, MapBalance picks a non-KoH layout. KGT broadcasts "[KGT] mod not active, spawn cfg missing" in chat 3 seconds after round start.

Capture math (per tick, default EvalIntervalSeconds = 1.0)

For every unit in Unit.Units: 1. **Inside zone?** $(dx^2 + dz^2) \leq \text{_captureRadius}^2$ (2D cylinder, Y unbounded) 2. **Eligible?** Skip if UnitCapValue ≤ 0 (harvesters, queens, etc.) 3. **Compute weight per unit:** `mult = Cfg.PerUnitOverride[name] ← e.g. Crab = 0.25 ??`
`Cfg.CategoryMultiplier[Cfg.BuildingToCategory[built_at]] ← Sol_Heavy → "HeavyVehicle" ?? 1.0`
`weight = UnitCapValue × mult if (player-controlled) weight ×= Cfg.PlayerControlledMultiplier`
(default 2.0) 4. **Accumulate per team:** `scores[team] += weight`

King ownership

After scoring:

```
total_in_zone = sum(scores.values())
for team in scores:
  if (scores[team] / total_in_zone) ≥ Cfg.DominanceThreshold (default 0.60):
    candidate_king = team
```

The candidate becomes the king after `Cfg.OwnershipHandoverDelay` seconds of sustained dominance (smoothing — prevents flicker when units flow through).

Accumulation toward win

Each tick the king accumulates:

```
this_tick_gain = min(scores[king], Cfg.MaxCapturePointsPerSecond × EvalIntervalSeconds)
king_progress += this_tick_gain
if (king_progress ≥ effective_threshold) → KGT TriggerWin
```

`effective_threshold` = `Cfg.WinThresholdPerMap[<scene>]` if present, else `Cfg.WinThreshold`.

If `Cfg.CumulativeAccumulation = true` (default), there's **one shared accumulator** — when a new team becomes king they continue from where the previous king left off. If false, each team has their own bucket.

Outpost ring fill: `kingN = round(OutpostCount × king_progress / threshold)` — the first N outposts (clockwise from north) show the king's faction colour; the rest stay neutral. Ring updates incrementally — only slots whose owner changes are despawned + respawned, so a +1 progress tick costs ≈ 2 networked ops instead of 24.

Capture-progress milestones

`king_progress / threshold` thresholds: 25%, 50%, 75%, 95%. Each crossing fires a chat broadcast + a milestone wav (`SoundMilestone25.wav`, etc.). At 100% the round ends — `kgt_win` log line, finishing-force units spawn (8/16/24 Goliath/Tank/SiegeTank around each losing critical), per-team win wav plays, then `GameMode.EndRound()` is called.

Reward systems

Three orthogonal credit sources running in parallel:

Source	Beneficiary	Formula	Modes
Kill reward	Killing player	<code>round(victim.cost × Cfg.RewardKillFraction)</code> per kill (units AND structures)	1, 2
Zone-presence reward	Player whose unit is in zone	<code>round(EvalIntervalSeconds × unit.cap × Cfg.RewardZonePresenceMultiplier)</code> per tick per controlled unit	1 only
Commander reward	Faction treasury	<code>round(min(team_score, MaxCapturePointsPerSecond × dt) × Cfg.CommanderRewardMultiplier × handicap_mult)</code> per tick per team. Goes via <code>Team.StoreResource</code> (overflow → <code>Team.StartingResources</code>)	1 only
Starter credits	Player	Flat <code>Cfg.StarterCreditsPerPlayer</code> granted once per round at game start (+ 5 s poll for late joiners)	1, 2

Costs honor live `Si_UnitBalance` `cost_mult` — KGT reads `Si_UnitBalance_Config.json` directly at each round start and multiplies the vanilla dump cost by the per-unit `cost_mult` field. (Reading

`ConstructionData.ResourceCost` would return vanilla because `UnitBalance`'s `OverrideManager` is a runtime overlay, not a field mutation.)

Commander-reward handicap (anti-snowball)

Every `HandicapRecomputeIntervalSeconds` (default 5 s):

```
score(team) = TotalResources × HandicapResourceWeight + Units.Count × 100 ×
HandicapMilitaryWeight
ratio       = score(team) / avg(score across non-neutral teams)
mult        = clamp( ratio^(-HandicapStrength), HandicapFloorMult, HandicapCeilingMult )
```

Defaults give: dominant team at $2\times$ avg $\rightarrow 0.50\times$ reward; trailing team at $0.5\times$ avg $\rightarrow 2.0\times$ reward. Floor `0.25`, ceiling `3.0` (live config currently `5.0`). Display per-team multipliers via `/koh status` (green when boost, red when handicap).

What MapBalance does for KGT specifically

1. **Layout selection bias** — `Si_MapBalance.IsKohModLoaded()` reflect-calls `Si_KingOfTheHill.KingOfTheHill.ModeKohActive`. When `true`, the layout picker prefers JSONs that carry a `koh` block. When `false` (Mode 2 / 3), the picker treats KGT as not loaded.
2. **Per-map spawn data** — `koh.PrefabName`, `koh.Position`, `koh.Team`, `koh.CaptureRadius`, `koh.ExclusionRadius` all live in the spawn-config JSON under `UserData/Spawns/<MapName>/koh_*.json`. Tune them per map.
3. **The publish handoff** — `MapBalance` writes `MapBalanceSpecials.Koh = { Obj, CaptureRadius, ExclusionRadius }` after spawning and fires `OnSpecialsReady`. KGT's only knowledge of *where* the capture point is comes from that public field.

To **disable KGT on a single map**, remove the `koh` block from that map's spawn-config JSON. KGT will broadcast the *"mod not active, spawn cfg missing"* notice and the round runs as normal Strategy.

To **change win threshold per map**, set `Cfg.WinThresholdPerMap` entries in `Si_KingOfTheHill_Config.json` keyed by Unity scene name (e.g. `"NorthPolarCap": 30000`).